

An aerial photograph of a university campus. In the foreground, a large, white, multi-story building with a red-tiled roof and arched windows is visible. The building is surrounded by green lawns and trees. In the background, a densely forested hill rises above the campus. The sky is clear and blue.

Baselining AI's Systems Engineering Performance in Cost Modeling

Boehm CSSE
Annual Research Review 2024
16-18 April 2024
Ryan Bell
Ryan Longshore
Raymond Madachy



Agenda

1. Guiding Research Question
2. Large Language Models
3. Benchmarks
4. Cost Modeling Benchmark
5. Implementation
6. Results and Discussion
7. Continued and Future Work



Guiding Research Question

How do we assess and compare performance of Large Language Models (LLMs) in the field of Systems Engineering, specifically cost modeling?

Large Language Models

- **Introduction to LLMs**

- Revolutionized NLP with human-like text generation.
- Trained on vast text data for wide topic understanding.
- Applications: Writing assistance, chat bots, code generation, summarization.

- **Types and Accessibility**

- **Open Source Models:** Llama 2, Falcon, Dolly.
 - Available on platforms like HuggingFace.
- **Proprietary Models:** GPT-4, Claude, Bard.
 - Generally outperform open-source models.



Hugging Face



Gemini

- **Training Sources**

- Diverse data sources: GitHub (code), wikis, textbooks, private corpuses.

- **Model Fidelity and Hardware Preferences**

- **Quantization Technique:** Runs inference on lower precision for local hardware ease.
- **Model Sizes Example:** Llama 2 available in 7B, 13B, and 70B variants.

- **Using Models: Prompt Engineering**

- **Zero Shot:** No context, single attempt.
- **Few Shot:** Multiple refining attempts.
- **Retrieval Augmented Generation (RAG):** Uses large data corpus for relevant info at inference.

- **Custom Fine-Tuning**

- Enhances performance on specialized tasks with domain-specific datasets.

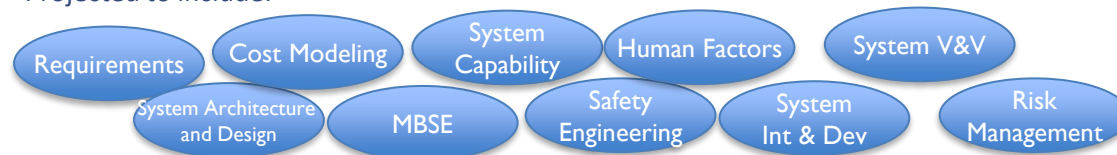


Benchmarks

- The landscape of AI benchmarks has evolved over time with early benchmarks focusing on foundational tasks such as word relationships and their semantic similarities to more recent, increasing complexity benchmarks such as College Medicine, Physics, Biology, Computer Science, Math, Electrical Engineering, among others
- We can see this progression with increased complexity and domain specific nature of the benchmarks over time
- Current benchmarks do not include system engineering
- A domain specific benchmark is needed

Enter SysEngBench

- A novel benchmark specifically designed to evaluate LLMs in the context of systems engineering concepts and applications.
- Projected to include:



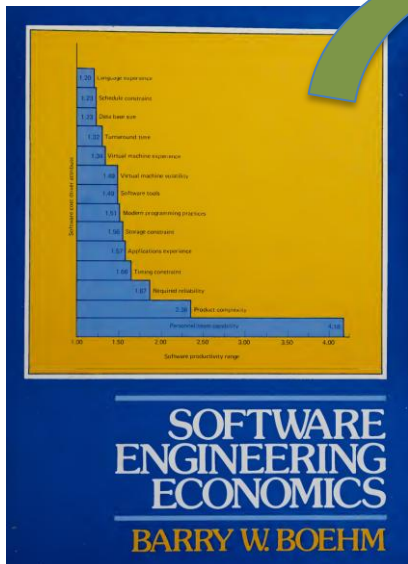
- Today is showing progress for the cost modeling section
- “By leveraging a diverse array of real-world and synthetically generated scenarios, SysEngBench aims to provide an assessment of LLMs' ability to interpret complex engineering problems and generate innovative solutions.”

Complexity and Domain Specific

Benchmark Name	Topic	Released
WordNet	Word relationships and meanings, foundational dataset for semantic similarity and language understanding	1985
MNIST	Handwritten digit recognition, foundational for image processing and computer vision	1998
BLEU	Language translation quality metric, foundational for evaluating machine translation systems	2002
ImageNet	Large-scale image recognition and classification, pivotal in advancing deep learning in computer vision	2009
LAMBADA	Understanding context and reasoning in text, focusing on predicting sentence endings (Paperno et al., 2016)	2016
SWAG	Common sense reasoning and predicting plausible sentence endings in a given context (Zellers et al., 2018)	2018
GLUE	A collection of diverse NLU tasks like question answering and sentiment analysis to advance language understanding across various contexts.	2018
SuperGLUE	A successor to GLUE with more challenging tasks, pushing the limits of NLU models with advanced reasoning and co-reference resolution.	2019
HellaSWAG	An extension of SWAG for more challenging common sense reasoning scenarios (Zellers et al., 2019)	2019
ARC	“ARC evaluates an AI's ability to tackle each task from scratch, using only the kind of prior knowledge about the world that humans naturally possess, known as core knowledge.” (Clark et al., 2018; Lab42, 2024)	2019
DROP	Reasoning over paragraphs, requires numerical reasoning and understanding of natural language (Dua et al., 2019)	2019
Winogrande	A large-scale dataset of winograd schemas designed to improve commonsense reasoning in AI systems.	2019
XTREME	Cross-lingual understanding and translation across multiple languages, tests multilingual capabilities	2020
MMLU	Measures professional and academic knowledge across various fields including College Medicine, Physics, Biology, Comp Sci, Math, Electrical Engineering, Professional Accounting, Psychology and worldly knowledge about Foreign Policy and Religions, among others (Hendrycks et al., 2021)	2021
GSM8K	Grade School Math 8K (GSM8K), a collection of math word problems aimed at evaluating numerical reasoning	2021
BIG-Bench	Broad spectrum of tasks testing reasoning, common sense, professional knowledge, and language capabilities (Google/BIG-Bench, 2021/2024)	2022



Cost Modeling Benchmark



Row Labels	Question Count	Question %
Cost Modeling	55	100.00%
Grand Total	55	100.00%

Source	Generati	Question	Choice A	Choice B	Choice C	Choice D	Answer	Justification
Software Engineerin	Synthetic	What approach does the text propose for software engin	Ignoring less q	The GOALS ap	Focusing excl	Outsourcing co	B	The GOALS approach is proposed as a method for integrating programr
Software Engineerin	Synthetic	Which of the following is NOT included as a cost driver in	Product compl	Software tool	Market demar	Required reliab	C	Market demand for the software is not listed among the cost drivers in
Software Engineerin	Synthetic	What does the COCOMO model primarily aim to estimat	The market va	The environm	The cost and s	The number of C	C	The COCOMO model is used for estimating the cost and schedule assoc
Software Engineerin	Synthetic	According to the sections, what role do software tools pl	They are irrele	They can pote	They solely in	They are only c	B	Software tools are considered as a cost driver in the COCOMO model, in
Software Engineerin	Synthetic	What does the Waterfall Model of software developme	Rapid prototy	Concurrent de	Sequential prc	Flexible respor	C	The Waterfall Model is defined by its strict sequential approach to soft
Software Engineerin	Synthetic	According to the text, what is a key economic rationale	It maximizes t	It minimizes r	It reduces cost	It accelerates d	B	The economic rationale for the Waterfall Model includes minimizing ris
Software Engineerin	Synthetic	Which of the following is NOT considered a refinement	Incorporating	Adding a phas	Integrating coi	Allowing for ph	C	While the Waterfall Model has been refined in various ways, including
Software Engineerin	Synthetic	What is a primary purpose of defining detailed life-cycle	To enable rapi	To facilitate cl	To simplify th	To ensure com	B	Detailed life-cycle phase definitions are critical for facilitating clear cor
Software Engineerin	Synthetic	The text describes the Software Work Breakdown Struct	To outline the	To detail the s	To provide a d	To track the sof	C	The Software Work Breakdown Structure (WBS) is designed to provide
Software Engineerin	Synthetic	In the context of software maintenance, what does the	The importanc	The necessity	The role of m	The need for a	C	The text underlines the role of software maintenance in extending the
Software Engineerin	Synthetic	What is the primary aim of the Basic COCOMO Model as i	To determine	To estimate th	To select the	To identify the	B	The Basic COCOMO Model aims to provide an empirical estimate of the
Software Engineerin	Synthetic	Which factor is NOT considered when defining the devel	Size of the sof	Number of de	Complexity of	Phase distribut	B	While the Basic COCOMO Model takes into account the size and comple
Software Engineerin	Synthetic	What role does the Rayleigh Distribution play in the Bas	It determines	It estimates th	It models the	It selects the	C	The Rayleigh Distribution is used within the Basic COCOMO Model to m
Software Engineerin	Synthetic	What does "Nominal Project Profiles" refer to in the con	The average p	Standardized	Default settin	Basic character	D	Nominal Project Profiles represent the basic characteristics or assumpt
Software Engineerin	Synthetic	How is interpolation used in the Basic COCOMO Model?	To adjust effor	To calculate ef	To determine	To select the m	B	Interpolation is used in the Basic COCOMO Model to estimate effort for
Software Engineerin	Synthetic	What is the primary focus of basic software maintenance	Predicting the	Estimating the	Calculating th	Determining th	C	Basic Software Maintenance Effort Estimation focuses on calculating th
Software Engineerin	Synthetic	What is a key aspect of the Basic COCOMO model's appr	It focuses on t	It estimates sc	It emphasizes	It considers the	B	The Basic COCOMO model primarily estimates software development e
Software Engineerin	Synthetic	Which factor is considered a significant cost driver in sof	The aesthetic	The geographi	Product comp	The color scher	C	Product complexity is recognized as a significant cost driver in software
Software Engineerin	Synthetic	In the context of software engineering economics, what	Agile develop	Waterfall dev	Prototype dev	Incremental de	B	While the Basic COCOMO model itself does not explicitly endorse a spe
Software Engineerin	Synthetic	What is emphasized as a crucial factor for improving soft	Strict adheren	The use of hig	The employm	Focusing on th	C	The employment of sophisticated software tools is emphasized as a cru
Software Engineerin	Synthetic	According to the content, what role does software proje	It is secondary	It is considere	It is crucial for	It is only releva	C	Software project planning and control is highlighted as crucial for ensur
Software Engineerin	Synthetic	How does the content describe the estimation of softwa	As a variable	As a fixed per	As dependent	As an integral	D	The estimation of software maintenance cost is described as an integra
Software Engineerin	Synthetic	According to the Basic COCOMO model, which of the foll	13 person-mor	55.76 person-r	145.93 person-	100 person-mo	B	For a project size of 20,000 DSI, the Basic COCOMO model estimates the
Software Engineerin	Synthetic	For the maintenance of a 10-KDSI organic-mode payroll	5 man-months	10 man-month	13.01 man-mo	15 man-month	C	For a project that involves adding 3000 DSI and modifying 2000 DSI annu
Software Engineerin	Synthetic	Which of the following options correctly lists the eight r	Analysis, Desig	Requirement	Planning, Anal	Conceptualizat	C	The software life cycle typically encompasses the following major phas
Software Engineerin	Synthetic	How does incremental development primarily impact so	It increases th	It leads to a m	It decreases o	It requires the	B	Boehm describes how incremental development modifies the traditior
Software Engineerin	Synthetic	What are the two main forms of advancement in software	Anticipatory d	Incremental d	User interface	Project manage	A	Advancement in the context of software development involves anti
Software Engineerin	Synthetic	In Boehm's analysis of present versus future expenditur	The diminishin	The importanc	The concept tl	The impact of c	C	Through the example comparing the rental versus purchase of process
Software Engineerin	Synthetic	What is the primary focus of performance models in the	Estimating the	Analyzing the	Predicting the	Understanding	B	Performance models in software engineering economics are tools used

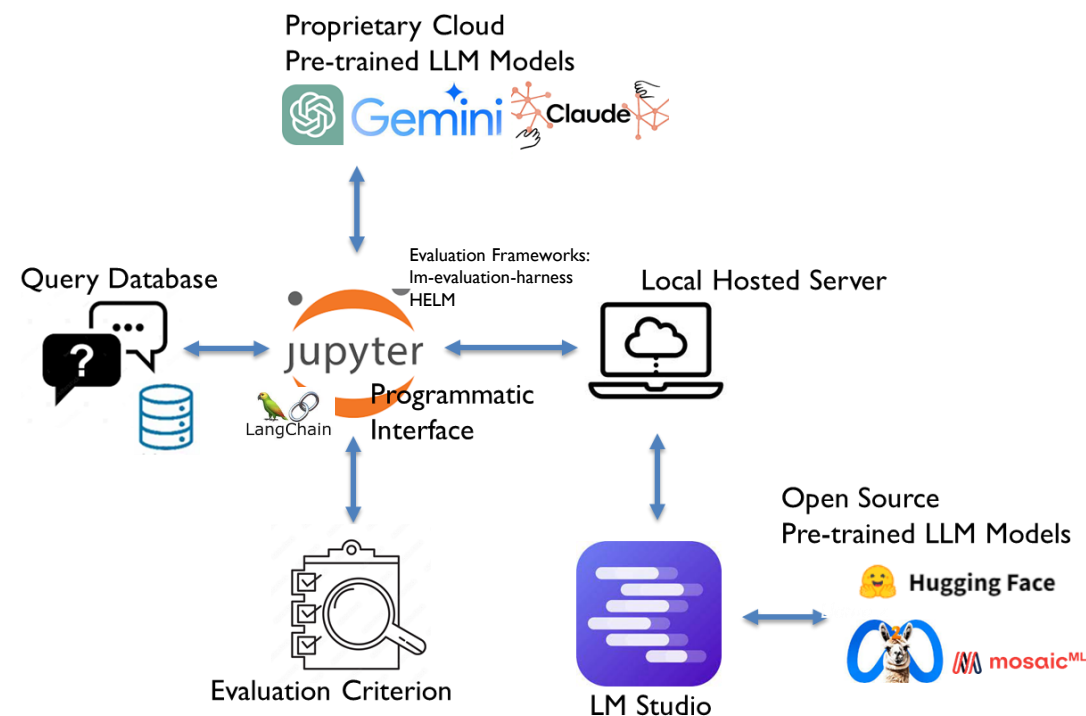
Implementation

- In order of increasing structure for Q&A:
 - Method 1: OpenAI simple Q&A chat completions
 - Method 2: Langchain
 - Method 3:
 - Langchain + AI harness
 - Langchain + HELM

Method 2

```

1 import os
2 import json
3
4 results_dict = {}
5 file_output_name = 'llm_output.json'
6 file_path = os.path.join(folder_directory, file_output_name)
7 # Loop through each row in the DataFrame
8 instructions = "You are taking a multiple choice test. Only provide your letter answer, no explanation"
9 for index, row in df.iterrows():
10     print("\nQuestion #" + str(row['Q#']) )
11     prompt = f"Question: {row['Question']}\nChoices:\nA. {row['Choice A']}\nB. {row['Choice B']}"
12     print(prompt)
13     result = chat([SystemMessage(content=instructions), HumanMessage(content=prompt)])
14     print(result.content)
15     # Use the question number (or any unique identifier) as the key
16     results_dict[row['Q#']] = result.content
17
18 # Write the results_dict to a JSON file in the specified directory
19 with open(file_path, 'w') as file:
20     json.dump(results_dict, file)
    
```





Results and Discussion

- Initial pass on the benchmark was ran for questions generated from “Software Engineering Economics” by Barry Boehm
- Questions were primarily simple conceptual questions
- Models without any advanced prompting techniques (e.g. RAG) or fine tuning performed fair, ranging from 65-73% correct
- Higher scores on conceptual portion may be needed for success with practical applications
- Need to investigate ability to answer application questions

		65.45%	72.73%	70.91%
Row Labels	Question Count	20240405 Llama 2 BCSSE	20240405 Mistral BCSSE	20240405 Orca 2 BCSSE
Cost Modeling	55	36	40	39
(blank)	55	36	40	39
Grand Total	55	36	40	39



Continued and Future Work

- Naval Postgraduate School Annual Acquisition Research (ARP) Symposium Presentation/Paper
 - “Introducing SysEngBench: A Novel Benchmark for Assessing Large Language Models in Systems Engineering”
- Future Work/Topics:
 - Complex Question Expansion
 - Subfield Diversification
 - Evaluation by Practicing Systems Engineers
 - Evaluation of Multiple Choice Question Bias within SysEngBench
 - Investigate the variance of choosing correct answers. Leverage the techniques performed by (Zheng et al., 2024).
 - Multimodal Input and Output
 - Investigate usage of images and diagrams both interpretation and creation for SE
 - Systems Engineering Domain Specific LLMs
 - Investigate custom LLMs for specific domains within systems engineering via RAG or fine-tuning. Compare domain specific performance against the SysEngBench.
 - Enabling Round Table AI Discussions for a LLM SE Team
- Other collaborations:
 - Ryan Longshore
 - Evaluation of LLMs with SysMLv2 Queries
 - Small Language Models for Domain Specific Knowledge
 - Raymond Madachy
 - Evaluation of LLMs for Modern Systems Engineering Cost Modeling with COSYSMO

An aerial photograph of a large, historic university campus. The main building is a long, white, multi-story structure with a red-tiled roof and numerous windows. To the right of the main building is a large, rectangular, blue swimming pool. The campus is surrounded by lush green lawns and trees. In the background, a densely forested hill rises above the campus, with some houses visible on the slopes. The sky is clear and blue.

Questions?

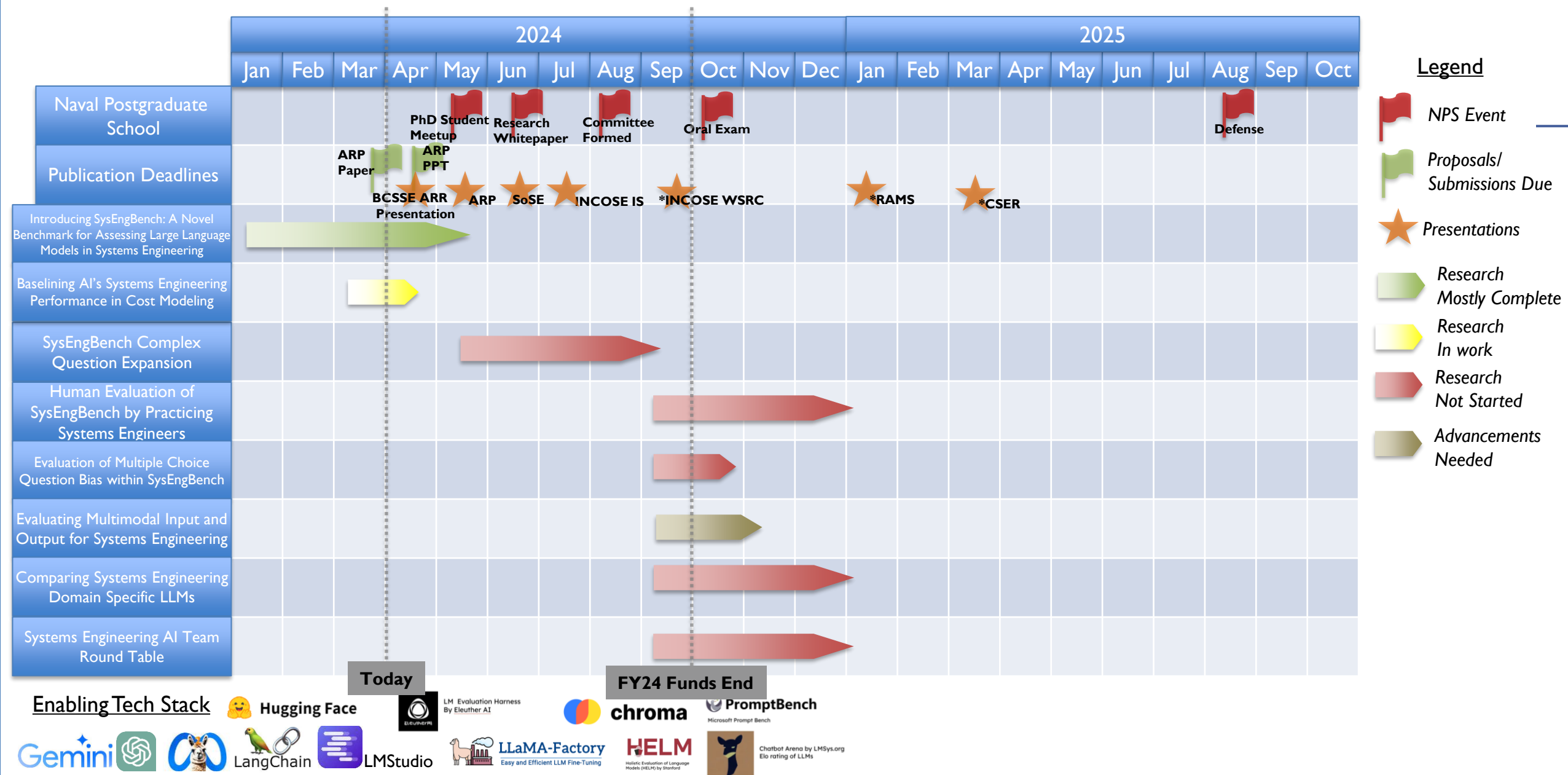


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Backup





Category	Sub-Category	Tags	Source	Generati	Question	Choice A	Choice B	Choice C	Choice D	Answer	Justification			
Cost Modeling			Software Engineerin	Synthetic	What approach does the text propose for software engin	Ignoring less q	The GOALS ap	Focusing excl	Outsourcing co	B	The GOALS approach is proposed as a method for integrating programm			
Cost Modeling			Software Engineerin	Synthetic	Which of the following is NOT included as a cost driver ir	Product compl	Software tool	Market demar	Required reliat	C	Market demand for the software is not listed among the cost drivers in			
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Cost Modeling			Software Engineerin	Synthetic	According to the sections, what role do software tools pl	They are irrele	They can pote	They solely in	They are only c	B	Software tools are considered as a cost driver in the COCOMO model, ir			
Cost Modeling			Software Engineerin	Synthetic	What does the Waterfall Model of software developmer	Rapid prototy	Concurrent de	Sequential pr	Flexible respo	C	The Waterfall Model is defined by its strict sequential approach to soft			
Cost Modeling			Software Engineerin	Synthetic	According to the text, what is a key economic rationale b	It maximizes t	It minimizes r	It reduces cos	It accelerates d	B	The economic rationale for the Waterfall Model includes minimizing ris			
Cost Modeling			Software Engineerin	Synthetic	Which of the following is NOT considered a refinement c	Incorporating	Adding a phas	Integrating co	Allowing for ph	C	While the Waterfall Model has been refined in various ways, including			
Cost Modeling			Software Engineerin	Synthetic	What is a primary purpose of defining detailed life-cycle	To enable rapi	To facilitate cl	To simplify th	To ensure com	B	Detailed life-cycle phase definitions are critical for facilitating clear con			
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Cost Modeling			Software Engineerin	Synthetic	In the context of software maintenance, what does the t	The importanc	The necessity	The role of m	The need for a	C	The text underlines the role of software maintenance in extending the			
Cost Modeling			Software Engineerin	Synthetic	What is the primary aim of the Basic COCOMO Model as i	To determine	To estimate th	To select the t	To identify the	B	The Basic COCOMO Model aims to provide an empirical estimate of the			
Cost Modeling			Software Engineerin	Synthetic	Which factor is NOT considered when defining the devel	Size of the sof	Number of de	Complexity of Phase distribut	B	B	While the Basic COCOMO Model takes into account the size and comple			
Cost Modeling			Software Engineerin	Synthetic	What role does the Rayleigh Distribution play in the Basi	It determines	It estimates th	It models the	It selects the o	C	The Rayleigh Distribution is used within the Basic COCOMO Model to m			
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Cost Modeling			Software Engineerin	Synthetic	What is the primary focus of basic software maintenance	Predicting the	Estimating the	Calculating th	Determining th	C	Basic Software Maintenance Effort Estimation focuses on calculating th			
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Cost Modeling			Software Engineerin	Synthetic	Which factor is considered a significant cost driver in sof	The aesthetic	The geographi	Product comp	The color scher	C	Product complexity is recognized as a significant cost driver in software			
Cost Modeling			Software Engineerin	Synthetic	In the context of software engineering economics, what	Agile develop	Waterfall dev	Prototype dev	Incremental de	B	While the Basic COCOMO model itself does not explicitly endorse a spe			
Cost Modeling			Software Engineerin	Synthetic	What is emphasized as a crucial factor for improving soft	Strict adheren	The use of hig	The employm	Focusing on th	C	The employment of sophisticated software tools is emphasized as a cru			
Cost Modeling			Software Engineerin	Synthetic	According to the content, what role does software proje	It is secondary	It is considere	It is crucial fo	It is only releva	C	Software project planning and control is highlighted as crucial for ensur			
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Cost Modeling			Software Engineerin	Synthetic	How does incremental development primarily impact so	It increases th	It leads to a m	It decreases o	It requires the	B	Boehm describes how incremental development modifies the traditior			
Cost Modeling			Software Engineerin	Synthetic	What are the two main forms of advancement in softwa	Anticipatory d	Incremental d	User interface	Project manage	A	Advancement in the context of software development involves anti			
Cost Modeling			Software Engineerin	Synthetic	In Boehm's analysis of present versus future expenditur	The diminishin	The importanc	The concept t	The impact of c	C	Through the example comparing the rental versus purchase of process			
Cost Modeling			Software Engineerin	Synthetic	What is the primary focus of performance models in the	Estimating the	Analyzing the	Predicting the	Understanding B	B	Performance models in software engineering economics are tools used			
Cost Modeling			Software Engineerin	Synthetic	How does Boehm describe achieving optimal performan	Maximizing so	Finding a bal	Prioritizing de	Focusing solely	B	Achieving optimal performance in software projects, as described by Bo			
Cost Modeling			Software Engineerin	Synthetic	What role does sensitivity analysis play in software engi	It identifies th	It determines	It analyzes ho	It focuses on m	C	Sensitivity analysis in the context of software engineering economics i			
Cost Modeling			Software Engineerin	Synthetic	What is the significance of cost-effectiveness models in	To calculate th	To determine	To predict sof	To solely focus	B	Cost-effectiveness models in software engineering economics are cruc			
Cost Modeling			Software Engineerin	Synthetic	What does Boehm suggest as a primary method to comb	Increasing the	Adopting mor	Breaking down	Leveraging adv	C	Boehm suggests that breaking down large software projects into small			
Cost Modeling			Software Engineerin	Synthetic	According to Boehm, how does the program size relate	t	Directly propo	Inversely prop	No direct rela	D	Boehm presents a case where the capacity to serve students per hour i			
Cost Modeling			Software Engineerin	Synthetic	What is gold-plating in the context of software engineer	The process of	The use of pre	Ensuring that	Applying rigor	A	Gold-plating refers to the practice of adding unnecessary features or er			
Cost Modeling			Software Engineerin	Synthetic	In the context of software engineering economics, what	Allocation is u	Allocation hel	Allocation is a	Allocation invo	D	In software engineering economics, allocation refers to the strategic di			
Cost Modeling			Software Engineerin	Synthetic	How do anticipatory documentation and software scaffo	They signific	They have neg	They potentia	They decrease	C	Anticipatory documentation and software scaffolding are proactive me			
Cost Modeling			Software Engineerin	Synthetic	In the economic decision-making process described by B	Because it ign	Because it ove	Because it fail	Because it doe	C	Boehm points out that evaluating the cost of renting versus purchasing			
Cost Modeling			Software Engineerin	Synthetic	What does marginal analysis primarily focus on in the co	The total cost	The increment	The initial buc	Long-term inve	B	Marginal analysis focuses on the incremental or marginal benefits and			
Cost Modeling			Software Engineerin	Synthetic	According to Boehm, which of the following factors signi	The physical si	The number o	The specific a	The programmi	C	Boehm suggests that the value of information processing products is si			
Cost Modeling			Software Engineerin	Synthetic	In the context of software engineering economics, how	c	It refers to the	It discounts fu	It calculates th	It is a method u	B	The concept of the present value of a series of cash flows is crucial in e		
Cost Modeling			Software Engineerin	Synthetic	Why is sensitivity to the interest rate or discount rate	sig	It influences t	It affects the	It determines	B	Sensitivity to the interest rate or discount rate is significant because it			
Cost Modeling			Software Engineerin	Synthetic	Which factor has the highest impact on software costs	ac	Language exp	Required relia	Virtual machi	r	Boehm's list of software cost driver attributes indicates that product co			
Cost Modeling			Software Engineerin	Synthetic	What is a critical observation Boehm makes about the CC	It includes all	It exclusively	It omits certai	It overemphasi	C	Boehm acknowledges that while the COCOMO model is comprehensive			
Cost Modeling			Software Engineerin	Synthetic	According to Boehm, how does software maintenance	fa	It is considere	It is the prima	Maintenance	C	Boehm places significant emphasis on the role of software maintenanc			
Cost Modeling			Software Engineerin	Synthetic	What is a critical consideration when choosing between	The aesthetic	The program	In-house vs.	v	The color scher	C	When deciding between developing a Special-Purpose Operating Syste		
Cost Modeling			Software Engineerin	Synthetic	What is the importance of presentation techniques for	u	They are cruci	They assist in	They eliminat	They primari	B	Presentation techniques for unquantifiable criteria are essential in sof		
Cost Modeling			Software Engineerin	Synthetic	In the context of software engineering economics, how	c	By focusing on	Through the u	Utilizing prese	Assigning arbit	C	Coping with unreconcilable and unquantifiable goals in software proje		
Cost Modeling			Software Engineerin	Synthetic	What strategy does Boehm suggest for dealing with deci	Always choose	Use statistical	Apply decisio	Ignore the unce	C	In the context of complete uncertainty, where probabilistic informatio			
Cost Modeling			Software Engineerin	Synthetic	According to Boehm, how does understanding the value	It eliminates	It provides a	It helps in det	It guarantees	t	C	Understanding the value of information allows project managers to ass		
Cost Modeling			Software Engineerin	Synthetic	In the realm of software engineering economics, what	rc	They dictate	They define	They help in	n	They are used	t	C	Utility functions are a tool used in economic analysis to model the pref
Cost Modeling			Software Engineerin	Synthetic	What is the first step in establishing objectives for softw	Determining	Identifying th	Outlining the	Calculating the	C	C	Establishing objectives in software cost estimation begins with outlinir		
Cost Modeling			Software Engineerin	Synthetic	In the context of software cost estimation, what is cruci	Ensuring unlin	Gathering hist	Prioritizing so	Securing additi	B	B	When planning for required data and resources in software cost estima		
Cost Modeling			Software Engineerin	Synthetic	Why is pinning down software requirements critical in th	It allows for th	It facilitates	It accurate requ	It helps in mark	C	C	Pinning down software requirements is critical because accurate and d		



Coding Framework

Desktop/NISE FY24 Risk LLM/ x Untitled - Jupyter Notebook x +

localhost:8888/notebooks/Desktop/NISE%20FY24%20Risk%20LLM/Untitled... x

UPDATE Read the migration plan to Notebook 7 to learn about the new features and the actions to take if you are using extensions - Please note that updating to Notebook 7 might break some of your extensions. Don't show anymore

jupyter Untitled Last Checkpoint: 8 minutes ago (unsaved changes) Logout

File Edit View Insert Cell Kernel Navigate Widgets Help Trusted Python 3 (ipykernel)

```
1 history = []
2 {"role": "system", "content": "You are an intelligent assistant. You always provide well-reasoned answers that are both correct and helpful."}
3 {"role": "user", "content": "Hello, introduce yourself to someone opening this program for the first time. Be concise."}
4 ]
5
6 while True:
7     completion = client.chat.completions.create(
8         model="local-model", # this field is currently unused
9         messages=history,
10         temperature=0.7,
11         stream=True,
12     )
13
14     new_message = {"role": "assistant", "content": ""}
15
16     for chunk in completion:
17         if chunk.choices[0].delta.content:
18             print(chunk.choices[0].delta.content, end="", flush=True)
19             new_message["content"] += chunk.choices[0].delta.content
20
21     history.append(new_message)
22
23     # Uncomment to see chat history
24     # import json
25     # gray_color = "\033[90m"
26     # reset_color = "\033[0m"
27     # print(f"{gray_color}\n{'-'*20} History dump {'-'*20}\n")
28     # print(json.dumps(history, indent=2))
29     # print(f"\n{'-'*55}\n{reset_color}")
30
31     print()
32     history.append({"role": "user", "content": input("> ")})
```

Hello! I'm Assistant, here to help you with any questions or tasks you may have. My purpose is to provide accurate and helpful information,

In []: 1

In []: 1

In []: 1

Model RAM Usage: 7.53 GB

TheBloke • llama 2 chat 7B Q8_0 gguf v

RAM Usage: 7.53 GB
CPU: 49.99 %

Local Inference Server

Start a local HTTP server that behaves like OpenAI's API. When running the server you will not be able to use the in-app Chat UI.

Configuration

Server Port: 1234

Cross-Origin-Resource-Sharing (CORS): ON OFF

Request Queuing: ON OFF

Verbose Server Logs: ON OFF

Apply Prompt Formatting: ON OFF

Start Server Stop Server

Client Code Examples

hello world (curl) chat (python) ai assistant (python)

```
if chunk.choices[0].delta.content:
    print(chunk.choices[0].delta.content, end="", flush=True)
    new_message["content"] += chunk.choices[0].delta.content

history.append(new_message)

# Uncomment to see chat history
# import json
# gray_color = "\033[90m"
# reset_color = "\033[0m"
# print(f"{gray_color}\n{'-'*20} History dump {'-'*20}\n")
# print(json.dumps(history, indent=2))
# print(f"\n{'-'*55}\n{reset_color}")

print()
history.append({"role": "user", "content": input("> ")})
```

Server logs

Server running on port 1234 (logs are saved into /tmp/lmstudio-server-log.txt)

Filter logs... Open Logs Clear (Ctrl+K)

```
[2024-02-13 13:19:40.079] [INFO] [LM STUDIO SERVER] Success! HTTP server listening on port 1234
[2024-02-13 13:19:40.079] [INFO] [LM STUDIO SERVER] Supported endpoints:
[2024-02-13 13:19:40.080] [INFO] [LM STUDIO SERVER] -> GET http://localhost:1234/v1/models
[2024-02-13 13:19:40.081] [INFO] [LM STUDIO SERVER] -> POST http://localhost:1234/v1/chat/completions
[2024-02-13 13:19:40.082] [INFO] [LM STUDIO SERVER] -> POST http://localhost:1234/v1/completions
[2024-02-13 13:19:40.084] [INFO] [LM STUDIO SERVER] Logs are saved into C:\tmp\lmstudio-server-log.txt
[2024-02-13 13:19:40.084] [INFO] [LM STUDIO SERVER] Setting context overflow policy to: Rolling Window
[2024-02-13 13:21:33.948] [INFO] [LM STUDIO SERVER] Processing queued request...
[2024-02-13 13:21:33.950] [INFO] Received POST request to /v1/chat/completions with body: {
  "messages": [
    {
      "role": "system",
      "content": "You are an intelligent assistant. You always provide well-reasoned answers that are both correct and helpful."
    },
    {
      "role": "user",
      "content": "Hello, introduce yourself to someone opening this program for the first time. Be concise."
    }
  ],
  "stream": true
}
```

Model Downloads

0 downloading • 2 completed

openhermes-2.5-mistral-7b-Q8_0.gguf
https://huggingface.co/TheBloke/OpenHermes-2.5-Mistral-7B-GGUF/resolve/main/openhermes-2.5-mistral-7b-Q8_0.gguf Finished Clear

llama-2-7b-chat-Q8_0.gguf
https://huggingface.co/TheBloke/Llama-2-7B-Chat-GGUF/resolve/main/llama-2-7b-chat-Q8_0.gguf Finished Clear



Script for Benchmark Evaluation

```

1 import os
2 import json
3
4 results_dict = {}
5 file_output_name = 'llm_output.json'
6 file_path = os.path.join(folder_directory, file_output_name)
7 # Loop through each row in the DataFrame
8 instructions = "You are taking a multiple choice test. Only provide your letter answer, no explanation."
9 for index, row in df.iterrows():
10     print("\nQuestion #" + str(row['Q#']) )
11     prompt = f"Question: {row['Question']}\nChoices:\nA. {row['Choice A']}\nB. {row['Choice B']}"
12     print(prompt)
13     result = chat([SystemMessage(content=instructions), HumanMessage(content=prompt)])
14     print(result.content)
15     # Use the question number (or any unique identifier) as the key
16     results_dict[row['Q#']] = result.content
17
18 # Write the results_dict to a JSON file in the specified directory
19 with open(file_path, 'w') as file:
20     json.dump(results_dict, file)

```

What best describes a system?

- A) A singular element operating independently
- B) A group of elements working together for a specific purpose
- C) A random collection of components without a defined objective
- D) An isolated technique without interaction with elements

Best choice:

B

Which of the following best encapsulates the basic methodology of Systems Engineering (SE)?

- A) Focus solely on delivering the solution quickly
- B) Understand the problem, examine alternatives, validate the solution, and deliver within constraints
- C) Start solving the problem immediately without understanding it
- D) Ignore alternative solutions and focus on the first idea

Best choice:

B

What are the core aspects of Systems Engineering (SE)?

- A) Focusing solely on the maintenance of simple systems
- B) Design, production, and maintenance of complex, multi-disciplinary systems within constraints
- C) Ignoring constraints such as cost and time
- D) Concentrating on a single discipline with no external constraints

Best choice:

A

In [15]: 1 gpt_answers

Out[15]: ['B', 'B', 'A']

In [25]: 1 grading_df = [df['Q#'], pd.Series(gpt_answers, df['Answer'])]

In [27]: 1 grading_df = pd.DataFrame({
2 'Q#': df['Q#'],
3 'GPT Answers': pd.Series(gpt_answers, index=df.index), # Ensuring index alignment
4 'Answer': df['Answer']
5 })
6

In [28]: 1 grading_df

Out[28]:

	Q#	GPT Answers	Answer
0	1	B	B
1	2	B	B
2	3	A	B